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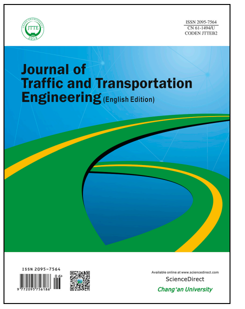

Recent advances in connected and automated vehicles

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Reviews

Recent advances in connected and automated vehicles

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Highlights

- This paper conducts a comprehensive review on recent advances of CAV technology.
- Five topics: communications, security, intersection navigation, collision avoidance, and pedestrian detection are included.
- These topics are essential to the success of CAV and need to be better understood by researchers and policy makers.
- Based on the current state-of-the-art, we provide insights on future research directions.

Abstract

Connected and automated vehicle (CAV) is a transformative technology that has great potential to change our daily life. Therefore, CAV related research has been advanced significantly in recent years. This paper does a comprehensive review on five selected subjects that lie in the heart of CAV research: (i) inter-CAV communications; (ii) security of CAVs; (iii) intersection control for CAVs; (iv) collision-free navigation of CAVs; and (v) pedestrian detection and protection. It is believed that these topics are essential to ensure the success of CAVs and need to be better understood. For inter-CAV communications, this paper focuses on both Dedicated Short Range Communications (DSRC) and the future 5G cellular technologies; for security of CAVs, this paper discusses both passive and active attacks and the existing solutions; for intersection control, this paper summarizes the pros and cons of both centralized and decentralized approaches; for collision avoidance, this paper concentrates on four subareas: maneuverability, vehicle networking, control confliction, and motorcycles; for pedestrian detection, this paper covers sensor, radar, and computer vision based approaches. Under each topic, this paper not only shows the state-of-the-art, but also unveils potential future research directions. By establishing connections among these subjects, this paper shows how they interact with each other and how they can be integrated into a seamless user experience. It is believed that the literature covered and conclusions drawn in this paper are very helpful to CAV researchers, application engineers, and policy makers.

Keywords:

Connected and automated vehicles; Autonomous vehicles; Intelligent transportation systems; Driver-less cars.

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1 Introduction

Connected and automated vehicles (CAVs) (a.k.a. connected and autonomous vehicles and driver-less cars) is a transformative technology that has great potential for reducing traffic accidents, enhancing quality-of-life, and improving the efficiency of transportation systems. Bajpai (2016) showcased the positive effects CAVs should have, compared to present experiences. For example, the reduction of distance in between vehicles for increased capacity but not raising delay times ends with a higher throughput. Other benefits may include declined emergency room patients, reduced car insurance premiums, and smaller sizes of traffic enforcement departments. Also, Bajpai noted that sharing a vehicle through third-party companies or individuals would lower the number of cars owned. The number of vehicles per household may drop with the availability of driver-less cars returning home to be used by other household members. Ride-sharing with autonomous vehicles can have a greater impact on available properties located within densely populated areas. Researchers (MRCagney, 2017) pointed out that individuals who did not have a license to drive would benefit greatly from CAVs by relying less on traditional forms of public transportation. The requirement for large parking areas will go away from rising use of autonomous car sharing that constantly scans for potential fares to pick-up. The constant hunting for riders will drive up the actual miles travel per vehicle compared to current standards. As mentioned, road incidents should be lower than present figures with the removal of human error in autonomous cars. It is estimated the traffic accidents will drop by 70% in 25 years (MRCagney, 2017).

Although the near future will be influenced by CAVs, current challenges continue to postpone the public implementation of this technology. The CAV uses wireless network and sensors to obtain relevant traffic and other vital information while its driving control is regulated by one of six levels of automation (SAE International, 2016). The levels range from zero to five: zero requiring complete human driving interaction and five being full autonomous navigation without human participation. As a wonderful asset to have in driver assistance, level 5 employs technology that relieves drivers from navigation responsibilities. The current level of automation found in today's vehicles is level 2, which consists of adaptive braking and acceleration, a technology that adjusts speed without driver assistance. Avoiding accidents with other vehicles or pedestrians has been a high priority topic, and once solutions have been determined and vetted, it will be a major hurdle conquered. Passenger and vehicle security remains a concern with evolving physical and cyber threat that may lead to personal information stolen to compromised CAVs being controlled by an unwanted user.

A main focus of this paper is to highlight the opportunities to reduce the traffic incidents that involve collisions especially with pedestrians or cyclists. A recent incident in Tempe, AZ, involved a human monitored CAV that struck a jaywalking pedestrian (The Verge, 2018). Since the incident, Uber has suspended the testing program. However, other companies like Google, Blackberry, and GM continue testing. Collision and pedestrian avoidance in conjunction with intersection control are extremely important to protect against injury or loss of life in the unfortunate event of an accident. Along with the protection of human life, the ability to communicate to the proper authorities when an incident has taken place is the next step in CAV evolution.

Many review articles are available in the literature for CAV related research. Bagloee et al. (Bagloee et al., 2016) create excellent reviews discussing the basic vocabulary and topics pertaining to CAVs in general terms, and Parkinson et al. (2017) explain topics in the cyber realm of CAVs with threats and solution tactics. While some review articles focus on less technical aspects of autonomous vehicles and more on comfort and personalization (Elbanhawi et al., 2015) and (Hasenjager and Wersing, 2017), some other ones cover specific and technical topics. In research of Deb et al. (2018), how pedestrians are willing to accept CAVs in the roadways is discussed. In research of Guanetti et al. (2018), the authors review the control aspects of CAV technology, where two topics are focused on: one is on-board real-time control, including subtopics such as powertrain control, motion control, and real-time motion planning; the other is remote planning and routing, including subtopics such as battery charge planning, eco-driving for both isolated and groups of CAVs, and eco-routing. The impact of CAVs and vehicular communications to traffic control is surveyed in research of Li et al. (2014). Both isolated intersection and network-wide traffic control are reviewed by the paper. The authors also discuss non-cooperative and cooperative driving under each category. Rios-Torres and Malikopoulos (2017) survey coordination of CAVs at intersections and merging at highway on-ramps. Both centralized and decentralized approaches are discussed, and within each category, the works are further classified into heuristic control and optimal control.

This survey paper is different from the ones above, and our contributions are two-fold. First, instead of concentrating on one or two topics, we conduct a comprehensive survey, focusing on five areas that lie in the heart of CAV research: inter-CAV communications, security and privacy, intersection navigation control, collision avoidance, and pedestrian detection. Second, we include the most recent research outcomes (within the past 2-3 years) in each covered research area. We believe that this survey paper provides sufficient breadth and depth, especially for researchers who are new to CAV research.

It is important to note that although we will present the five research topics in different sections, they are not isolated and independent from each other. As we will see later in the paper, these CAV research areas are highly correlated and must be evaluated altogether in real-world experimentations. For example, inter-CAV communications is the enabling technology for enhancing efficiency and safety of CAV; without reliable inter-CAV communications, it is going to be extremely hard to achieve intersection control and collision avoidance. Similarly, security plays a key role in CAV research. If a CAV is compromised by hackers, then all the safety features are in jeopardy. Finally, the design of collision avoidance and pedestrian detection need to involve social considerations. For instance, the answer to the following question is very debatable: in case of an unavoidable accident that caused by the sudden appear of a pedestrian on the street, should the CAV choose to hit the pedestrian or another object on the street? The former choice would hurt the pedestrian while the latter may actually injure the passengers in the CAV. Of course, such an extreme scenario may not happen very often in reality, but it further indicates that in order to ensure the successfulness of CAVs, it is important to raise the awareness of all related research topics and understand how they interact with each other.

The remainder of this paper will be divided into the following parts: Section 2 discusses some of the applications and technologies found in inter-CAV communications; Section 3 explores the use of and prevention of security attacks susceptible to the CAV environment; Section 4 breaks down the logic behind centralized and decentralized intersection navigation control; Section 5 reviews various works dedicated to maneuverability, networking, and control confliction that are implemented to ensure safe avoidance of traffic incidents with other vehicles; Section 6 dives into the technology and methods imperative to protect pedestrians and other users of adjacent areas to roadways used by CAVs; Section 7 concludes with suggestions to future opportunities.

2. Inter-CAV communications

Drivers navigate roadways and intersections by applying the information located on roadside signs, on radio stations, or other forms gathered before entering the vehicle for travel. The combination of all that information is necessary to make the correct choice when a situation arises. CAVs require a similar stream of data to institute a proper protocol for roadway events. The ways these vehicles obtain that information is slightly different to the human counterpart. For example, vehicle communication allows for vehicles to relay information between one another about various environmental changes such as a large influx of traffic (Fang et al., 2017). The NHSTA predicts that effectively applying vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications could potentially reduce and/or eliminate up to 80% crashes of any type from non-impairment (NHTSA, 2017). Fang et al. make note of two other forms of vehicle communication included in intelligent transportation systems: vehicle-to-pedestrian (V2P) and vehicle-to-network (V2N). All these communication mechanisms are collectively known as vehicle-to-everything (V2X).

V2X heavily relies on wireless technologies. Dedicated short range communications (DSRC) is a type of wireless technology developed for the automotive platform, specifically for CAV usage. Specifically, DSRC uses radio frequencies that lie in a range of the 5.9 GHz band, which is controlled and allocated by the Federal Communications Commission (FCC). DSRC is the pathway that allows CAVs to communicate with each other and the infrastructure. In terms of communication range, DSRC covers a maximum of 500 feet in all weather conditions (Lee and Lim, 2013). IEEE 802.11p, the IEEE standard for DSRC, has been implemented and endorsed by various automotive manufacturers and the U.S. Department of Transportation. There are multiple layers, as laid out by Kenney (2011), that are contained inside the DSRC platform: physical layer, data link layer, middle layers, and message sublayer.

DSRC system is broken down into two categories of hardware: the on-board unit (OBU), a.k.a. on-board equipment (OBE), and the road side unit (RSU) (Fang et al., 2017). As the name states, the OBU is stationed within the vehicle and operates as a multichannel transceiver for transmitting and receiving signals, while the RSU interchanges information with the OBU from elsewhere. The RSU can be stationed anywhere from another vehicle to various infrastructures. An additional difference between the OBU and the RSU is that the latter can only function when stationary, while the former can operate either stationary or mobile. Fig. 1 shows how information is exchanged between OBUs and RSUs. Due to the centralization of the data collected by OBUs, DSRC can be one of leading reasons why CAVs will help the traffic congestion caused by the human facet of driving (NHTSA, 2016). V2V networking has started to find its place in CAV accident prevention, but it will be discussed further in the collision avoidance section. DSRC is one of the central reasons behind the gradual increase of implementing V2V communications into collision avoidance. In research of Huang and Lin (2014), vector-based cooperative collision warning system (VCCW), one component of the proposed setup places the vehicle into a network of other CAVs where each vehicle periodically updates its position with other vehicles using DSRC. The concept focuses on intersections and curves, two of the hardest aspects for CAVs collision avoidance to overcome, but they make note of DSRC related works that are more applicable to the highway facet of collision avoidance (Huang and Lin, 2014; Young et al., 2008, 2009).

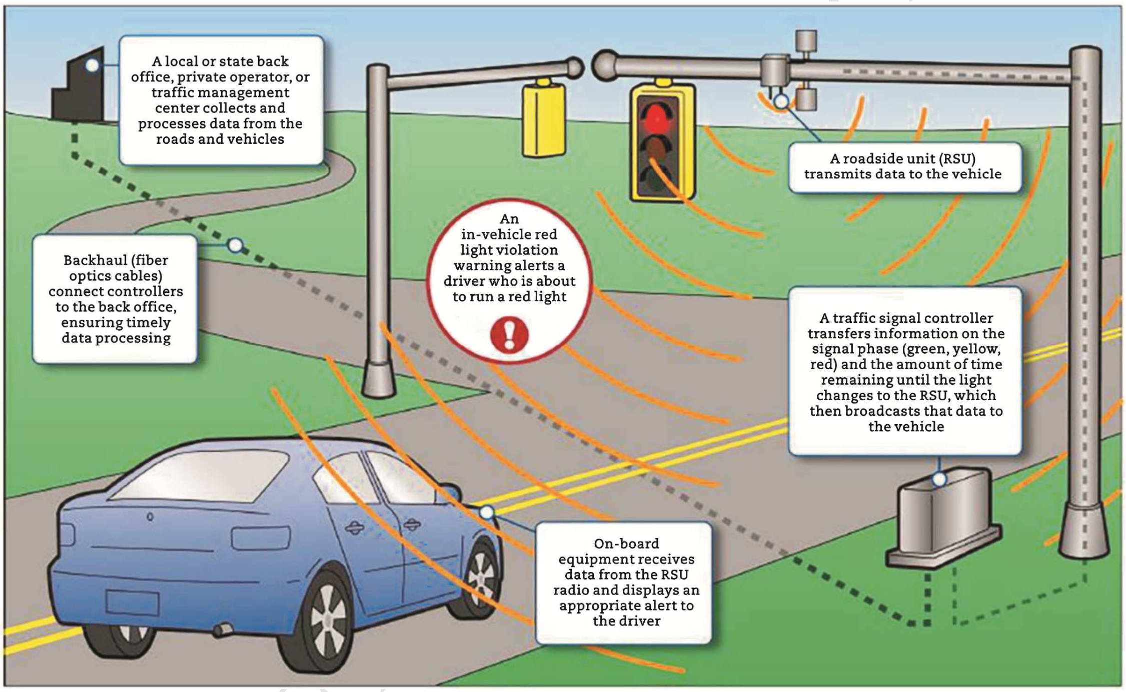

Fig. 1 DSRC architecture (United States Government Accountability Office Report to Congressional Requesters, 2015).

Through research and experiments, V2X can be expanded past the four primary categories to include additional technologies and objects. Choi et al. (2017) design their work around vehicle-to-railroad (V2R) communication and how it can be used as an additional warning system. The setup can be broken down into two different formats: direct and indirect. In the direct case, the warning transmission is sent directly to the vehicle. In the indirect case, the warning is sent first to the RSU, which then sends the transmission to the OBU. Each case performs optimally in different scenarios: the direct approach is for lower vehicle and locomotive speed and the indirect case is for higher vehicle and locomotive speed. Choi et al. tested an omni-directional and bi-directional antenna. The bi-directional antenna proved to be advantageous in urban areas but performed ineffectively in the rural environment. The counterpart was successful in the rural environment but underachieved in the urban environment. Railroad crossings are often looked over when it comes to both non-CAVs and CAVs. Nevertheless, they pose a larger threat to vehicle safety if taken lightly. With the constant growth of populations and density in major cities, it becomes harder for DSRC to be as effective in the benefits that they can provide, e.g., limits invisibility and blind spots (Biddlestone et al., 2012). The Center for Automotive Research from Ohio State University developed a traffic simulator called the Vehicle and Traffic Simulator (VaTSim), which they began testing in 2012 with the help of Honda R&D Americas Inc. (Biddlestone et al., 2012). In this paper, Biddlestone et al. reveal the overall performance of the simulator with DSRC produced accurate results and can be exploited as a method to test DSRC further.

The Shapely-value game theory applies a concept that materials or rewards are distributed based on the role a participant plays in cooperative action. Shah et al. (2018) explore the use of Shapely-value for an adaptive transmit power cooperative congestion control (AC3) to reduce the control channel congestion in situations with a high number of CAVs in a region. In current methods, vehicles adjust the transmission power by the level of saturation of the control channel with reduction in channel access; greater the saturation inversely lower the transmit power. Shah et al. offers creating a "coalition" of CAVs, during times of high congestion events, to impartially reduce transmit power by the contribution of each CAV, thereby reducing the overall load of the system without sacrificing the coverage for all participants.

Lorenzen (2017) offers a modification to linear message rate integrated control (LIMERIC) to assist in congestion control of the DSRC network. LIMERIC is a message rate congestion control that allows for a robust stability and convergence in a noiseless environment to adjust the messaging rate of each CAV (Kenney et al., 2011). However, Lorenzen offers that LIMERIC is designed to be an unlimited self-regulating system, and it falls short by being too aggressive and compromising the stability of the system. Self-weighted rate control (SWeRC) is the modification that allows for a fair message rate of convergence. Adjusting the weighting of the convergence parameter β reduces oscillating of message rates by using the previous normalized rate as the maximum achievable rate (Lorenzen, 2017). The weighting calculation is shown below.

$$\omega = \frac{r_{\mathrm{c}}(t-1)}{K}\frac{1}{R_{\max}T_{\mathrm{xtime}}} = r_{\mathrm{t}}(t-1)c \tag{1}$$

where ω is the weighting for LIMERIC control, $r_{\mathrm{c}}(t-1)$ is the total channel utilization of the previous iteration, c is the maximum local node capacity at the maximum rate $R_{\max}$ considering the transmission delay T_{xtime}, K is the number of nodes, and r_{t} is the group rate that converges to the target utilization of resource allocations. The theory was proven in testing, but it was limited to stationary nodes at close range.

The intention of DSRC is to be a path for V2X communication, but it has the ability to be applied into another use, range estimation, which is completed by expensive LIDAR or RADAR sensors a majority of the time. Niesen et al. (2017) propose the use of periodic broadcast ranging (PBR) to determine the distance between two CAVs. The method is based on the fact that a wireless signal's propagation delay is proportional to the distance traveled from a transmitter to the receiver. The research also has assumptions that safety messages are sent intermittently in the DSRC standard with an estimated time period of T = 0.1 s and the CAV internal clocks are unsynchronized. Vehicles s and t transmit a safety message with a timestamp located within the message, and a timestamp is created when the safety message is received. These timestamps, referred to as “departing" and “arriving", permit the calculating of the true synchronized time in between the two CAVs. Once time is synchronized, the range is estimated by the travel time in between the vehicles. The estimation of distance is calculated by the equations below.

$$\Delta s_{\mathrm{A}}(n) = (1+\delta)\,\Delta t_{\mathrm{D}}(n) + (d_{\mathrm{D}}(n) - d_{\mathrm{D}}(n\text{-}1))/c + \Delta z_{\mathrm{A}}(n) \tag{2}$$

$$\Delta s_{\mathrm{D}}(n) = (1+\delta)\,\Delta t_{\mathrm{A}}(n) + (d_{\mathrm{D}}(n) + d_{\mathrm{A}}(n\text{-}1))/c + \Delta z_{\mathrm{D}}(n) \tag{3}$$

with

$$\Delta z_{\mathrm{A}}(n) \triangleq z_{\mathrm{A}}(n) - z_{\mathrm{A}}(n\text{-}1) \tag{4}$$

$$\Delta z_{\mathrm{D}}(n) \triangleq z_{\mathrm{A}}(n) - z_{\mathrm{D}}(n\text{-}1) \tag{5}$$

where n is the nth message, Δs_{A}, Δs_{D}, Δt_{A}, and Δt_{D} are the timestamp differences for vehicles s and t in departing (D) and arriving (A) messages, respectively, δ is clock drift, c is the speed of light, and z is receiver noise (Niesen et al., 2017).

One significant drawback of DSRC powered V2X communications is its low scalability. When the number of nearby vehicles is high, it is unable to provide satisfactory time-probabilistic characteristics (Campolo et al., 2011; Van Eenennaam et al., 2009). In addition, DSRC is mostly designed for quick transmission of short-range basic safety messages and does not offer high bandwidth and low latency communication channels for more advanced V2X applications such as autonomous driving.

An alternative to DSRC is cellular based V2X (C-V2X) communications. However, 3GPP LTE, a 4G V2X solution, a.k.a., release 14 C-V2X, is incapable of providing high throughput and low latency for advanced applications. In study of Vinel (2012), Vinel develops analytical frameworks to compare the performance of DSRC and 3GPP LTE in terms of the probability to deliver the safety beacon before the expiration of the deadline. Numerical results show that when 50 vehicles are present, the probability of 3GPP LTE is actually worse than that of DSRC, which is at 83% and already below the requirements in typical safety application. What is really promising, and future proof is the 5G C-V2X, i.e., release 15 and 16 C-V2X. Compared with DSRC and 3GPP LTE, 5G C-V2X offers very higher throughput and reliability (99.999%), longer range (443 m line of sight and 107 m none line of sight), and very lower latency (10 ms end-to-end and 1 ms over-the-air) (5G Americas, 2018). In addition, 5G C-V2X provides direct messaging services among CAVs, allowing short-range communication when cellular towers are unavailable. Like other generations of cellular services, 5G requires years of development and evolution. The standard of 5G C-V2X was completed in March 2017, and the real products will be available after 5G becomes mature after 2020. The roadmap of 5G can be found in IEEE report (IEEE 5G, 2017).

DSRC is a vital component in the environment of CAV safety and data transfer. A constant and dependable wireless signal that extends past findings of 500 feet (Lee and Lim, 2013) allows for larger participation and receipt of safety messages. When the number of CAVs that are occupying an area increase greatly, the demand of communication congestion control rises. Shah et al. (2018) and Lorenzen (2017) are some of the examples needed to moderate the demand of wireless systems in high usage events. DSRC applications will integrate more into the everyday use of CAVs and passengers, while congestion control and coverage range are essential in road safety.

While DSRC is mostly targeted for basic safety applications, 5G C-V2X will enable more advanced CAV features such as autonomous driving and highway platoon. Compared with DSRC, 5G C-V2X has significantly increased throughput and reliability and reduced latency. Currently, 5G C-V2X products are being developed, and it is important to test, verify, and improve the performance when they become available. Mathematical models are also needed to compare the performance of

5G C-V2X and DSRC in vehicular applications. In addition, it is expected that the two wireless communication technologies will co-exist in the foreseeable future. Therefore, it is mandatory that we understand the pros and cons of using both signals. Real-world testbeds are needed to test co-existence and backward compatibilities under not only normal conditions, but also inclement weather and complex terrain. Finally, new network control methods are needed so that even if certain wireless links' quality is degraded severely due to congestion and/or interference, the overall effectiveness of the inter-CAV network is not significantly reduced so as not to cause catastrophic disasters.

3. Security of CAV

CAVs, along with any other computing platform, are susceptible to two forms of attack, passive and active. Passive attacks read information that is being transferred between a CAV and another communication point, e.g., RSU and a different CAV. This threat type has lower risk compared to active attacks where the results may end fatally. Active attacks may consist of spoofing incorrect data, resending a previous message to obtain validated system keys, message modification of relevant data, or denying of service that prevents data transfer on an affected server where data transference is vital.

3.1. Passive attacks

A CAV will broadcast a message containing verified velocity, location, a pseudonym, i.e., V_a or V_b, for the car, and other information to alert other vehicles nearby for safety purposes. Attackers may eavesdrop location information and car pseudonyms for initial selection of target car. Verified data, similar to broadcast safety messages, is transferred to an RSU through a virtual machine (VM) (Yu et al., 2013). VMs are secure cloud connections to local RSUs and the CAV. The data from the VM gives the attacker a second observable data group. In Fig. 2, Kang et al. (2016) explained a method that obtaining unique and precise information from a roadside location unit (RLU) about a target car can observe a tie between VM and V_a. As long as the VM is unrevised through the exchanges from one RLU to another or among different locations, the attacker is allowed to collect location data continually.

This threat could be modified to the second type of attack if the VM identity and the vehicle pseudonym used are not changed at the same time. An attacker can link VM identity and the pseudonym, if either one has a name modification for security and the other remains the same. If an attacker is monitoring a target car's location data during the asynchronous change, the unaltered parameter, for instance a vehicle pseudonym or the assigned VM, becomes a constant identifier during the change, referred to as linkage mapping attack (Kang et al., 2016).

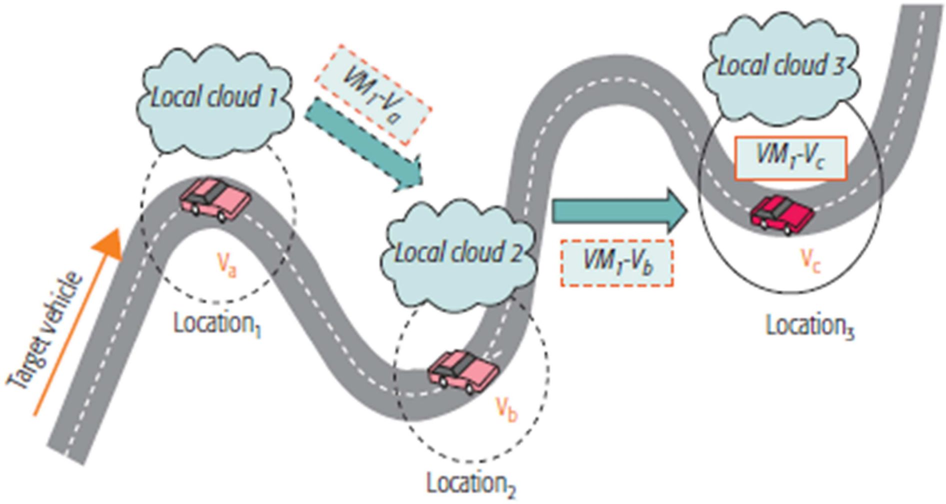

Fig. 2 Observation mapping attack (Kang et al., 2016).

Fig. 3 displays that even though the VM identity is changed at time interval, T_1, the pseudonym remains V_a in a separate time interval T_2.

Kang et al. (2016) offers solutions in these types of attacks. Observation mapping mitigation relies on applying a one-time use of VM identifiers. Vehicles register to the local network, applying for randomized VM identifiers from a VM manager. Through an encrypted connection, the VM manager assigns a store of VM identifiers to use for every location request attempt. Once the attribute is used by that specific vehicle, it will not be reused until it has been reassigned by the VM manager in another vehicle. So repeated use of an attribute cannot be tracked by simple observation. Linkage mapping attack prevention employs only synchronous pseudonym and identifier changes. The main part of the attack is based on asynchronous application of vehicle tags. The vehicle requests a pseudonym/identifier change, the two machines determine and confirm a window for receiving and application of new tag. At specified time, T, the vehicle and VM manager apply the change to both pseudonym and VM identifier. That synchronized change removes the traceable memory of the target vehicle (Kang et al., 2016).

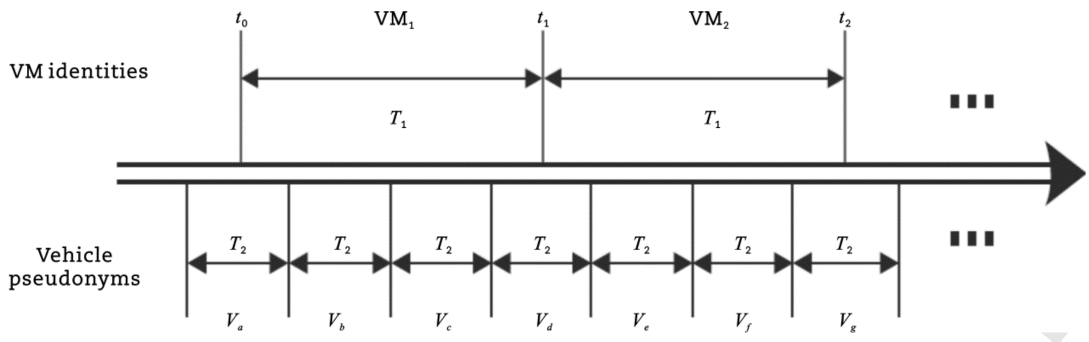

Fig. 3 Linkage mapping attack (Kang et al., 2016)

3.2. Active attacks

At roadway level, GPS signals are susceptible to malicious jamming attempts because their power levels are low after long distance attenuation. Using a jamming signal as an active attack, the GPS can be overpowered and rendered ineffective. The danger comes in the inability to determine when and where to stop without the locating signal. In an attempt to reduce the effectiveness of jamming, Ozdemir and Aksoy (2017) combine adaptive null-steering with antenna arrays to determine a direction of origin for unwanted multiple jamming signals. Once direction of the signals is estimated, a waveform needs to be produced for counteracting the jamming. Using an orthogonal array optimization method conceived by Taguchi (Ozdemir and Aksoy, 2017) the system compares an incoming jamming signal to a proposed broadcast cancellation signal using a fitness function.

$$f(x) = 2\sum_{n=1}^{4}(a_n \cos(\frac{2n-1}{2})kd\cos(\theta)) \tag{6}$$

$$\text{Fitness} = \int_{0^\circ}^{180^\circ} |\,\text{AF}_{\text{desired}}(\theta) - \text{AF}_{\text{obtained}}(\theta)\,|\,\mathrm{d}\theta \tag{7}$$

where AF is the array function for optimization, a_n is the value looking to be optimized for correct radiation pattern, k is the wave number, d is the half-wavelength element spacing, $\text{AF}_{\text{desired}}\,(\theta)$ is the side lobe level limited, null placed pattern, and $\text{AF}_{\text{obtained}}\,(\theta)$ is the radiation pattern obtained from AF equation (Ozdemir and Aksoy, 2017). As the integrated number moves closer to zero, the fitness function value represents a better-quality signal waveform match and thus reducing the effect of the jamming signal.

An active attack may involve the intelligent transportation system as well as CAVs. Changing the distance in between autonomous vehicles can disrupt the flow of traffic and increase chances of accidents. Ghanavati et al. (2017) determined through partial differential equations and Lyapunov functions that by subtly modifying velocities between malicious and normal vehicles, the spatial rules that control distance between the two cars will attempt to correct and lead to shock waves of traffic flow, traffic jams, etc. The logic is anisotropic, meaning that CAVs weigh the actions of vehicles in front significantly higher than ones located behind. This attack method does not involve connection of software or hardware located on CAVs, but exclusively uses operating rules for distance control to manipulate velocity and density of CAVs in a given section of roadway. Targeted vehicles are manipulated into a predetermined speed and capacity profile for the attack to work effectively. Once at the desired parameters, malicious vehicles subtly reduce speed to trigger the distance control functions on the vehicles located behind. The smaller in velocity alteration, the greater the effectiveness of the attack is due to the difficulty of detection.

Victim of a Global Navigation Satellite System (GNSS) spoofing attack is misled by illegitimate broadcast signals mistakenly treated as trustworthy. Spoofers can jam the original GNSS signal, causing the receiver to look for and acquire the incorrect signal instead of the proper signal. A more effective spoof is to gradually increase the amplitude of the illegitimate signal, A_{spoof}, until the amplitude of the real signal, AGNSS, is overpowered while attempting to maintain a Doppler-shift near to the correct GNSS signal to avoid detection. The third possible spoofing attack uses a null effect, in which, two spoofing signals are broadcasted: the first in a conventional spoofing method, while the second signal is the negative phase (180 degrees out of the proper GNSS signal phase), in effect canceling that signal out (Psiaki and Humphreys, 2016). A method of detecting spoofing attempts is using the automatic gain control (AGC) located on the front-end components of the GNSS receiver. Akos (2012) shows in efforts to spoof the GNSS signal, the total power levels of signals coming into the GNSS receiver increases. The GNSS receiver employs an AGC to control the constant power level that inputs into the analog-to-digital converter. Signal levels incoming to the AGC are inverted compared with the antenna, meaning that when power levels are low at the antenna, the AGC needs additional power to amplify the signal to the desired output level. When the AGC power draw is dropping, that is a good indicator of a spoofing attempt due to the additional signal levels added by the attack (Akos, 2012).

Han et al. (2017) develop a detection and mitigation method for spoofing using carrier frequency information and code delays. This method works successfully when the spoofing signals are more powerful than the authentic GNSS signals. When used in conjunction with other detection methods, low or equal power attacks are still detected. Their method can be implemented in two modules, centralized and distributed spoofing-mitigation.

The denial of service (DoS) attack is a detrimental process that produces false messages to the tipping point where the infected system appears to be unavailable or non-functioning. In the CAV platform, this attack is especially dangerous. Malina et al. (2015) use group verification with seven state process to remove harmful users on the communication channels. The seven stages are: setup, registration, join, signing, categorized verification, trace, and revocation.

Biron et al. (2018) determined a real-time detection of a DoS attack on cooper active adaptive cruise control (CACC) with considerations for vehicle platooning. They point out that most DoS attacks are packet loss or time delays, and they focus on the delay as the marker for the intrusion. They create the dynamics for the detection the time delay between the vehicles by using the following equations (Biron et al., 2018).

$$\dot{\hat{v}}_i(t) = L_v sgn\left(v_i(t) - \hat{v}_i(t)\right) \tag{8}$$

$$\begin{aligned}\dot{\hat{a}}_i(t) = \tfrac{k_p}{h} d_i(t) - \left(k_p + \tfrac{k_d}{h}\right)\hat{v}_i(t) - \left(k_p + \tfrac{1}{h}\right)\eta(t) \\ + \frac{k_d}{h} v_{i-1}(t) + \frac{1}{h}\left(a_{i-1}(t) - \dot{a}_{i-1}(t)\hat{\tau}\right) \\ + L_a\big(\eta(t) - \hat{a}_i(t)\big)\end{aligned} \tag{9}$$

$$\dot{\hat{\tau}}(t) = -\tfrac{L_b}{h}\dot{a}_{i-1}(t)\left(\eta(t) - \hat{a}_i(t)\right) \tag{10}$$

where $\hat{v}_i(t)$ and $\hat{a}_i(t)$ are the estimated relative velocity and acceleration, respectively; k_p and k_d present controller gains; h is headway time; $\hat{\tau}(t)$ is the estimated delay; L_v, L_a, L_b are the constant detector gains to be designed; $v_{i-1}(t)$ and $a_{i-1}(t)$ on-board vehicle sensor measurements for vehicle $i-1$; $d_i(t)$ and $v_i(t)$ are radar measurement from vehicle $i-1$; $\eta(t)$ filtered version of the signal $L_v\mathrm{sgn}\left(v_i(\mathrm{t}) - \hat{v}_i(t)\right)$ (Biron et al., 2018). Radar measurement noise, un-modeled dynamics, and underlying delays in communications were not considered with the dynamics equations. To combat this, a probability for false alarms was constructed.

$$P_{\mathrm{FA}} = \int_{\delta}^{\infty} p_0(x)\mathrm{d}x \tag{11}$$

where δ is the selected threshold for DoS attack and $p_0(x)$ is the $\hat{\tau}$ probability distribution under no DoS attack in the dedicated short-range communications (Biron et al., 2018).

The management of the vehicular Ad-Hoc networks (VANETs) consists of an approved authority (AA) and local general managers (GM). The AA creates pseudonyms and elliptic curve digital signature algorithm (ECDSA) key pairs for use on the network. Individuals sign up for a verified driver profile and register each vehicle with the AA. This solution is designed for a non-autonomous vehicle environment, but it also applies to CAVs because passengers of CAVs can attack and manipulate the network. AA processes the individual's name and scans the global revocation list (GRL) (Malina et al., 2015) for a hit. The GRL contains all banned users and vehicles. The user/vehicle has verified certificate from AA and attempts to join the VANET by a second verification with the local network GM. After gaining access to the VANET, users/vehicles may broadcast messages throughout the local network. Categorized verification works at the single vehicle level or the group level. At the group level, verification can accommodate n messages (Malina et al., 2015) use the small exponents test to check broadcasted messages, randomly.

$$\prod_{j=1}^{n} R_{j3}^{\theta_j} = e\left(\prod_{j=1}^{n}\left(T_{j3}^{S_{jx}} h^{-S_{j\delta}-S_{j\mu}} g_1^{-C_j}\right)^{\theta_j}, g_2\right) e\left(\prod_{j=1}^{n}\left(T_{j3}^{C_j} h^{-S_{j\alpha}-S_{j\beta}}\right)^{\theta_j}, \omega\right) \tag{12}$$

and if

$$1_{G_1} = \left(R_{j5}R_{j2}\right)^{-\theta_j} T_{j2}^{\theta_j S_{jx}-\theta_j C_j} \upsilon^{(S_{j\beta}-S_{j\mu})\theta_j} \tag{13}$$

where T is time stamp, S is different component of the signature, j is component from the product of the group signature bilinear map, x is element of the group secret key, υ is element of the group public key, C is the hash value in the group signature, G_1 are multiplicative cyclic group of a prime order of a temporary pairing, e and θ are random elements. If these two equations prove true, then the signed message is valid. Invalid messages are flagged: GMs determine which users the messages are originated from and added their name to the GRL with the approval of AA.

The layer where sensors interact with onboard computers on all vehicles is controller area network (CAN), and it designates the location of data payloads. An attack could use the CAN to intercept, modify, and inject compromised data for gaining control of a specific system or complete car. A properly structured malicious data packet is problematic to detect because that packet consists of valid messages, even though dangerous, located in proper data fields and transmitted at the recognized timing interval. The assessment of a CAN attack is challenging since vehicle manufacturers use different data packet protocols in brands, years, or models. It is difficult to program and detect an attack designed for a protocol that only works for one-year model. Taylor et al. (2018) use anomaly detection powered by efficient recurrent neural networks (RNNs) (Wang et al., 2017), where previous information, words, or calculations are utilized to identify the next malicious activity on CAN bus. Recorded CAN transmissions of a driving session is used first to teach the RNN what to look for. Specifically, the group (Taylor et al.,2018) perceived word sequences on the CAN bus mimicking the structure of a vector-space model. The RNN can extrapolate data packets similarly to words of the vector-space model. The derived difference in predicted data and actual data creates an anomaly signal by using a log loss function shown below.

$$L(\hat{b}_{n,m}, b_{n,m}) = -\left(b_{n,m}\log(\hat{b}_{n,m} + \epsilon) + (1 - b_{n,m})\log(1 - \hat{b}_{n,m} + \epsilon)\right) \tag{14}$$

where $b_{n,m}$ is the mth bit in packet p_n, $\hat{b}_{n,m}$ is the mth bit's estimated value, and ϵ is a fixed value to limit the maximum loss.

Moore et al. (2017) work on a solution for the regular-frequency signal injection attacks, in which an adversary repeatedly injects CAN bus signals with fixed process ID (PID) and at a rate greater than or equal to the expected rate of signal. The solution is based on the observation that the most PIDs' signals are regularly occurring, and the inter-signal arrival time never differs more than a 24% from expectation. Specifically, the authors exploit the relative regularity of the signals by using the percent of absolute error of an observed wait time from expectation, which is obtained from a few minutes of training when there is no active attack. To verify the effectiveness of the proposed method, they injected three types of attacks using an Arduino board connected to a vehicle via the OBD-II port. The first two attacks manipulate the runtime lights when the engine is on/off, and the third one forces the car's engine to turn off. Experimental data obtained from real-world CAN bus signals indicate that the true positive and false discovery rate of the method is 0.99998 and 0.00298, respectively.

To prevent security attacks, Satam et al. (2017) have sought out a manner for developers to use a systematic approach to include security programming during creation of entertainment or information services, instead of including the security measures after producing the desired service of music, destination routing, or other applications. The auto information development framework (AIDF) in Fig. 4 comprises of end devices, communications, services, and applications. Information that transfers throughout the system is subject to a passive or active attack. The auto security development framework (ASDF) employs multi-level intrusion detection to flag suspected activity by comparing it to normal system usage. A decision fusion engine combines all the flagged content for authentication along with reducing unnecessary handling of false alarms. Feedback from the decision fusion engine allows the ASDF to take actions on possible attack surfaces to mitigate attempt.

There have been various passive and active attacks on CAVs. Although solutions to these existing security threats are available, it is important to continue to keep investigating on possible new threats and improving solutions of current security systems found in CAVs.

One area that needs improvement is the false discover rate of security attacks on the CAN bus. Although the result of 0.00298 in study of Moore et al. (2017) is quite impressive, it may still cause quite some CAVs to be mistakenly classified as under attack, given the fact that many CAVs will be running for an extended period.

5G C-V2X security is another important research area. Since 5G C-V2X products are still under development, the related security attacks will not start until at least a couple of years later. The good news is that the being-developed 5G security features are well defined and heavily rely on authentication and encryption. Nonetheless, its effectiveness needs to be tested thoroughly before and after CAVs are deployed in the field, and the security protocols may need to be updated later.

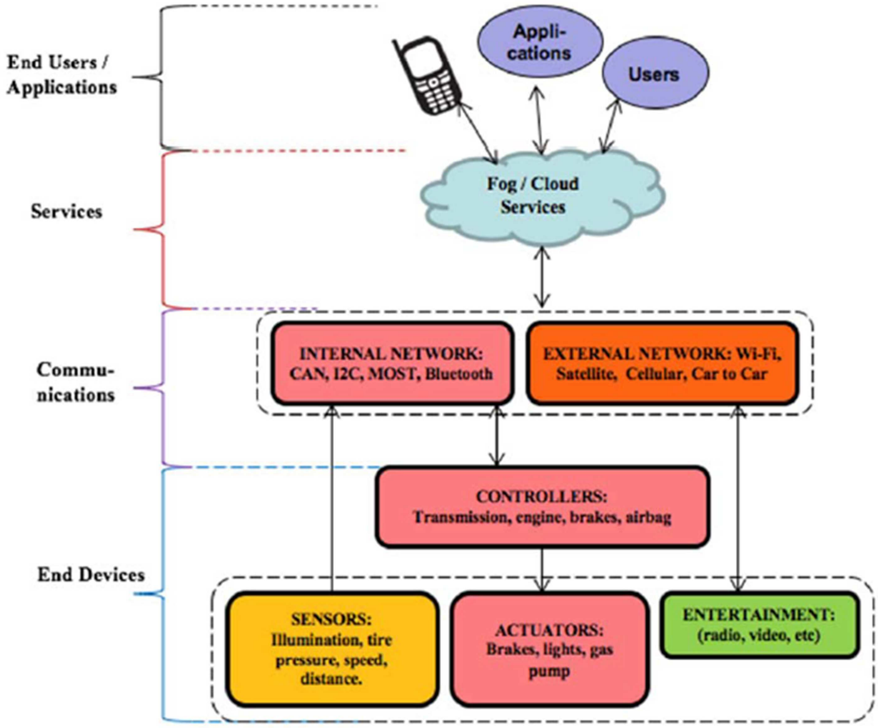

Fig. 4 Auto information development framework (Satam et al., 2017).

Regardless of the types of security threats, they can be equated to an organism: they can be stopped for a period of time but may evolve into a different threat by finding a different manipulation.

In addition, human factors of making mistakes give hackers opportunities to exploit viable development frameworks in vetting malicious programs. Given the fact that CAVs will not be operated by drivers, we believe that artificial intelligence and machine learning algorithms are needed to automatically detect security problems with no or minimal human intervention. In addition, CAVs should have basic rules built-in so that they will never cause crashes and hurt pedestrians on purpose. This is extremely important in the worst and extreme case that some CAVs are fully compromised.

4. Traffic intersection navigation

The requirement for traffic lights at intersection diminishes when CAVs take over the roadways. In this section, we discuss both “centralized” and “de-centralized” intersection control approaches. The difference between the two is that the infrastructure makes decisions in the former methods whereas the CAVs make decisions in the latter.

4.1. Centralized navigation

Li and Wang (2006) propose a centralized cooperative driving mechanism for an isolated blind crossing. Their goal is to achieve both safety and efficiency using inter-vehicle communications, which is done by dividing vehicles within a virtual circle centered at the junction into groups of maximize size 3. To ensure collision free driving, the authors come up with the concept of safe driving patterns, which are essentially vehicle pairs that can pass the intersection simultaneously without the risk of colliding into each other. A driving schedule can then be represented as an ordered series of safe patterns. The authors develop the basic solution tree generation algorithm to generate the possible driving schedules. By using the first-in-first-out property of traffic flow, the modified solution tree generation algorithm is proposed by the authors to prune the solution tree. Furthermore, the authors use the safety pair labeling algorithm to prune the unsafe driving schedules from the solution tree. To come up with the best cooperative driving schedule and vehicle trajectory, the authors divide the driving scenarios into four cases and derive an optimization problem for each case. Finally, the best driving plan is obtained by comparing all feasible solutions and finding the one that yields the least amount of time. The method proposed in the paper is verified by simulation. The authors discuss several future directions in the paper, including combating the complexity of the centralized approach, improving the tree-based algorithms for multiple-lane crossings cases, incorporating real-world driving scenarios, and relaxing some of the assumptions such as knowledge of perfect vehicle speed and position.

Dresner and Stone (2008) suggested the control of an isolated traffic intersection via CAVs communicating with an autonomous agent. A reservation-based controller is proposed to either grant or reject requested reservations based on an intersection control policy. The authors show that the proposed approach can dramatically reduce the average travel delay compared with a traffic signal controller. They also consider some practical factors in their experimentations including 5% packet loss in communications and combination with current traffic signal controllers to allow both human drivers and CAVs to use the system.

Zohdy et al. (2012) control CAV trajectories using cooperative adaptive cruise control (CACC) systems to avoid collisions and minimize delays at an isolated intersection. They assume that the followings are known to the controller: physical characteristics of the CAVs, entry speed and acceleration of the vehicles, weather condition (wet or dry), and intersection characteristics (number of lanes, lane width, etc.). To minimize the amount of time the vehicles use the intersection, the authors assume that the CAVs pass through the intersection at speed limit, i.e., the highest speed. To this end, the authors propose a two-zone intersection model (shown in Fig. 5): in Zone I, vehicles accelerate to maximum speed and maintain that speed until the end of the designated length; in Zone II, CAVs may change their speeds, but the maximum speed should be maintained in the end of the zone. Assuming that all CAVs traverse the intersection at full speed, the authors simplify the optimization problem, and the only control variable is the vehicle arrival time at the intersection stop bar. By defining the optimal time (OT) to be the shortest time possible to reach the stop line (i.e., no deceleration in Zone II), the objective of the optimization problem is to minimize the sum of the differences between the actual time (AT) and OT while the constraints include the first-come-first-served rule in the same lane and no conflict among vehicles in the intersection box. To make the controller as close to real-world scenarios as possible, a vehicle dynamics model and the Virginia Tech Comprehensive Power-based Fuel Model (VT-CPFM) are used. The proposed method is suitable for receding horizon control, and simulation results indicate that the savings in delay and fuel consumption are in the range of 91 and 82 percent, respectively, relative to conventional fixed-cycle signal control. In future work, the authors wish to include more types of vehicles and incorporate manually driven cars, pedestrian movements, etc. into the system model.

Lee and Park (2012) develop a centralized cooperative vehicle intersection control (CVIC) algorithm for CAV only scenarios at an isolated intersection without turning movements. The goal is to design the algorithm to manipulate individual vehicle's maneuvers so that the vehicles can safely cross the intersection without collisions. The paper assumes perfect inter-CAV communications, level terrain, and zero friction between the tire and the ground. By studying the vehicle dynamics and the trajectories, the authors derive an optimization problem in which the objective is to minimize the total length of overlapped trajectories subject to three constraints: (1) maximum acceleration or deceleration rate; (2) maximum and minimum speeds; and (3) minimum headways between two consecutive vehicles in the same lane.

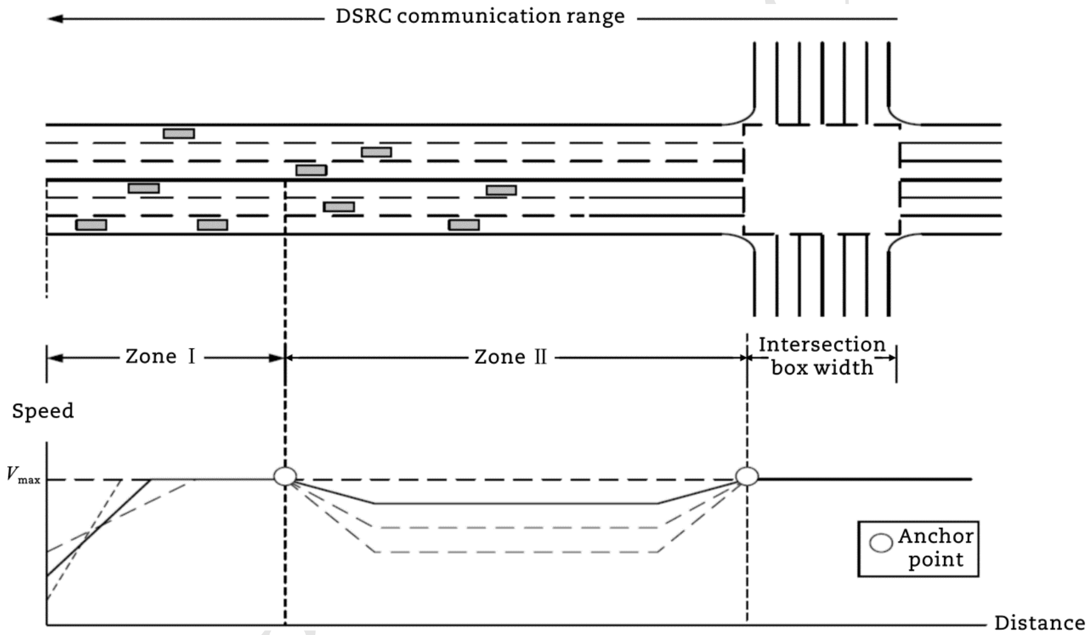

Fig. 5 Intersection and zone models of iCACC (Zohdy et al., 2012).

To maximize the chance of finding the optimal solution to the derived nonlinear constrained programming problem, three different methods provided by MATLAB: active set method based on sequential quadratic programming, the interior point method, and the genetic algorithm are used by the authors. In case that the problem is infeasible, the system enters a rule-based recovery mode in order to ensure safety. The performance of the CVIC algorithm and the controller is verified by simulation using the VISSIM simulator. The results show that compared with the actuated intersection control, 99% and 33% of stop delay and total travel time reductions, respectively, are achieved. In addition, the authors use the VT-Micro model in the simulation to investigate the environmental impacts. It turns out that the CVIC algorithm contributes to 44% reductions of CO_2

emission and 44% savings of fuel consumption. The CVIC algorithm is extended in study of Lee et al. (2013) for a corridor consisting of multiple intersections.

Similar to what they achieve in studies of Dresner and Stone (2008), Huang et al. (2012) come up with a more advanced reservation-based intersection controller for CAVs. Their contributions are multi-fold. First, they develop an integrated simulation testbed with four independent layers: traffic simulation, communication, infrastructure, and analyzer. The encapsulation property makes the testbed very flexible. For example, the communication layer may be integrated with a network simulator, such as ns-3 or NetSim, in order to simulate complex and real-world communication environments. Second, the new reservation algorithm recommends a speed profile for vehicles to follow until they leave the intersection and also permits assigning different priorities to incoming reservation requests. Third, the integrated simulator evaluates not only mobility (e.g., average wait time), but also environmental (e.g., fuel and emission) benefits. Simulation results show that the proposed approach can reduce the vehicle delay by 85%, fuel consumption by 50%, and emissions by 39%-50%. Note that the integrated simulation in study of Huang et al. (2012) can also be used to simulate multiple interconnected intersections.

Wu et al. (2013) develop a sequence-based framework, known as cooperative vehicle-actuator system, for isolated intersections. Both CAVs and manually driven vehicles are accommodated by the system, where the goal is to find the optimal passing sequence of all the vehicles. Two different control strategies are proposed. In the first approach, a receding horizon controller that assumes full knowledge about the cars in a limited time window is used to solve an optimization problem, in which the objective is to find the optimal sequence that minimizes the maximum exit time of the last vehicle subject to safe headway constraints. It is shown by the authors that dynamic programming can dramatically reduce the time complexity when the number of lanes is fixed. In the second method, a petri net model is used for two-way intersections, and some simple and local rules that lead to a global complex behavior are dis covered. Microscopic simulations are used to show that both control strategies work equally well and can dramatically reduce average delay compared with other approaches. The authors also run experiments at a physical intersection to prove the feasibility of the system in real-world applications.

Kamal et al. (2015) study a centralized CAV coordination problem at isolated intersections using model predictive control (MPC) and nonlinear programming. The scheme ensures safety by preventing each pair of conflicting vehicles from approaching their cross-collision point (CCP); it also utilizes the intersection more efficiently because not the whole intersection is reserved at once. In the MPC framework, a constrained nonlinear programming problem (NLP) is solved over a finite horizon to derive the optimal control sequences of the vehicles. Specifically, the objective function of the NLP is

$$J = \sum_{t=0}^{T-1}\sum_{i=1}^{N} w_{v_i}(v_i(t+1) - v_{\mathrm{d}})^2 + \sum_{t=0}^{T-1}\sum_{i=1}^{N} w_u\big(u_i(t)\big)^2 + \sum_{t=0}^{T-1}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N} F_{i,j}(t) \tag{15}$$

where T is the length (in steps) of the prediction horizon, v_{d} is the desired velocity, and w_{v_i} and w_u are weight coefficients, which are for quite computationally demanding, and heuristics are often needed to return the (near) optimal solutions quickly.

Zhang et al. (2015) develop a state-driven priority scheduling mechanism for CAVs passing through isolated intersections. Some terminologies in computing such as critical section and priority inversion are applied to intersections and CAVs. The authors first define critical sections of intersections and then come up with detailed parameters of various traffic objects, including lane and path, critical sections, vehicle model, and reservation-oriented vehicle action model. The authors then derive the state transition diagram and several rules, using which a priority scheduling algorithm known as sPriorFIFO is developed and verified by simulation.

Wuthishuwong and Traechtler (2017) offered a bi-level control: intersection and network level for intersection safety. This control is using the autonomous agent, referred to as autonomous intersection management (AIM). Intersection level transfers acceptable routes and speeds via vehicle to infrastructure communication, while network level employs infrastructure to infrastructure to control throughput and intersection balancing at a neighborhood level permitting the quickest and most efficient routes of travel. The AIM determines the neighborhood traffic density by polling all requested messages from incoming CAVs. The traffic density and optimal velocity through is calculated by the intersection manager using the consensus algorithm using the incoming and outgoing vehicle numbers in other networked intersections.

Lin et al. (2017) explain a methodology that an intersection manager, referred to as connected vehicle center (CVC), controls conflict sections with buffer assignments. Buffer assignments are like reservation-based control that was introduced by Dresner and Stone (2008) .Conflict sections are the points where two paths through an intersection bisect each other and may be locations of traffic incidents. Lin et al. proposed using two separate stop lines, one for CAVs and the other for human drivers, to help manage the intersection. The first stop line would regulate by visual traffic lights for the human drivers, and further back would lie the second stop line that CAVs use to determine when to enter the intersection. If there is a failure to calculate a crossing opportunity by the CVC, the CAV will stop at the first stop line, which is closer to the intersection, and wait for a special crossing time interval and a three-stage linear speed profile: accelerating, intermediate, and decelerating.

In a preassigned or reservation arranged bisection plan, proper vehicle spacing is required to ensure safe navigation of the intersection. Chai et al. (2017) employ four safety states of the distance between CAVs that are shown in Fig. 6: invalid, failure, valid, and small value. Invalid and valid are straight forward descriptions with invalid not having the required distance for complete passage, and valid slot finishes the crossing while maintaining the suggested velocity profile from the intersection manager. A failure slot is a condition where a distance between crossing vehicles permits traveling through the slot but at a velocity profile that exceeds the posted speed limits of the intersection. Finally, a small value slot is a length that allows a car to cross but accelerating quickly and not exceeding the posted speed limit. The intersection manager determines the category of each vehicle by using the length of the car, velocity profile (which can be adjusted), and the number of vehicles interacting within the intersection. They define two methods: location optimization on sequence evaluation (LOOSE) and cooperative optimization method for previous allocation alternatively transforming (COMPACT). LOOSE is the methodology that enforces proper vehicle spacing, and COMPACT distributes velocity profiles for intersection navigation.

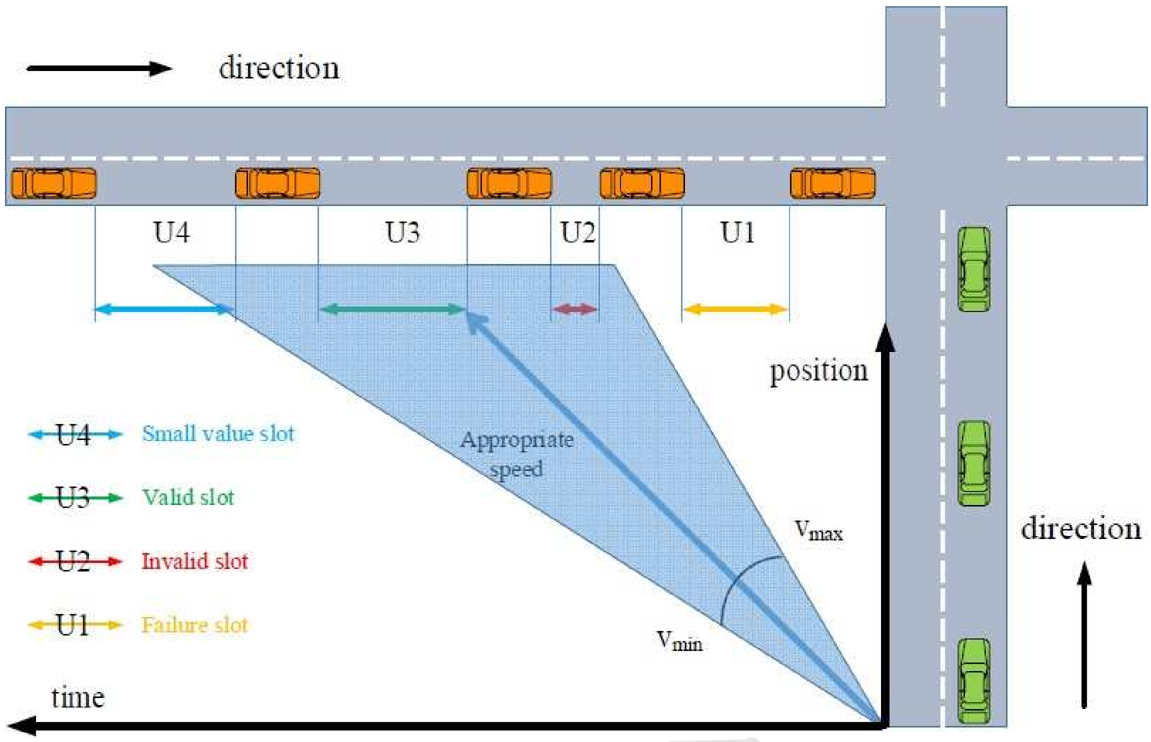

Fig. 6 Chai's slot types (Chai et al., 2017).

Another reservation-based method is submitted by Levin et al. (2017) where the use of an integer program to optimize the selection of conflict region tiles in an intersection allows for a reduction in total travel time. The tile-based reservation method lends itself to smaller intersections for flow rate improvements. They allow simultaneous control of an intersection tile in which two vehicles are turning in conflicting movements if the time step allows. Time steps are the divisions of time allocated for a vehicle to move through the intersection completely.

Levin and Rey (2017) offer a modification to autonomous intersection management in the application of their new mixed integer linear program (MILP). The intersection manager offers a reservation order placement to the incoming CAV, but that offering may be accepted or rejected. If the placement is turned down, a penalty is implemented to the CAV in an additional delay by requiring the new reservation request. This method emphasizes the acceptance of the initial reservation placement. Levin and Rey present an objective function found in their MILP that may adjust the reservations to balanced and fair state to all users of the intersection.

$$\min \sum_{i \epsilon \mathcal{V}} f_i\left(t_i(\gamma_i^+) + \tau_i(\gamma_i^+)\right) \tag{16}$$

where i is CAV labeled i, $\mathcal{V}$ is the set of vehicles, f_i is the priority of each CAV, and $t_i(\gamma_i^+) + \tau_i(\gamma_i^+)$ is the time CAV i leaves the intersection. Combining the objective function and MILP, Levin and Rey coin the name conflict-point intersection control (CPIC), which outperforms first come, first served (FCFS) policy by reducing total delay times of intersections with densities of 450 vehicles per hour from over 1000 s down to just over 200 s.

A guaranteed hull methodology relies on the paths of all vehicles are accounted for. Colombo et al. (2017) build a control mechanism using guaranteed hull principles that use clusters or platoons to bisect an intersection. The intersection manager builds impromptu vehicle groups to reduce the number of conflicts point reservations. This mechanism may be susceptible to reduced throughput if cluster sizes are not optimized depending on roadway occupancy. Wu et al. (2017) also employ a methodology that gives preference to platoons (vehicle groups) in certain situations. An instance of the methodology would be an intersection having two CAVs headed eastbound with estimated entrance to the intersection at 3 and 7 s. In a northbound lane, six CAVs have an intersection arrival time of 4 to 6.5 s. The rolling time window algorithm offered by Wu et al. permits the platoon to be split into two smaller groups while allowing most of the platoon to bisect the intersection on the second step after the CAV with the arrival time of 3 s.

Intersection control includes the velocity profile of cars, permitting entrance into the intersection, and so on. There are participants of intersections that do not directly connect with the manager. Pedestrians and cyclists use intersections in similar manners to CAVs, however they are considered higher priority in most cases. It is the responsibility of the manager to allow intersection crossings of these groups. Sarkar et al. (2017) propose a path mapping process that uses acquired data from a camera mounted on an intersection to develop cluster patterns of vehicles' paths.

The process uses the camera stills to determine velocity and inferred path direction (Sarkar et al., 2017). These patterns are used to create what they call an "influence network." The network applies learned patterns to incorporate pedestrian and cyclist crossing paths.

4.2. Decentralized navigation

An earlier effort of decentralized navigation can be found in study of Milanés et al. (2010), where the authors use a CAV and a manually driven car to do intersection control. Both cars are mass-produced vehicles, equipped with differential global positioning system (DGPS), PCs, and Wi-Fi adapters. The major difference between the two vehicles is that the former's actuators, i.e., steering wheel, throttle, and brake, are modified for autonomous driving. The goal of the work is to control the CAV based on the information of the other manually driven car to let the two vehicles pass an intersection collision free. The control system contains two parts. In the first part, the CAV uses V2V communications to detect the other vehicle's location, velocity, and direction when it is within 80 m; the point of intersection is then estimated based on both cars' trajectories. In the second part, a rule based fuzzy controller converts the input fuzzy variables to actuator control via a three-step process: fuzzification, inference engine, and defuzzification. Experimentation results show that based on the real-time V2V data, the CAV can either yield to the manually driven car or traverse the crossroad before it while ensuring safety.

Alonso et al. (2011) proposed two methods of determining the traffic intersection queue. Priority charts are the first of Alonso's group suggestion. The CAV will connect to a database inquiring about an upcoming intersection and obtain characteristics of potentially connected vehicles that may alter the intersection navigation timing by causing a stop or deceleration. They suggest using an occupancy vector in which the intentions of each vehicle through the intersection is logged in a range of 0-3 with counter-clockwise labeling of CAVs. We created Fig. 7 to help demonstrate the vector numbering from the reference vehicle as (1,3,2,0), where 0 is no vehicle, 1 is a right turn, 2 is straight through, and 3 is a left turn. Yield, stop, and no entry signs are handled in a similar vector with the values being: (Y) for yield, (S) for stop, (N) for no entry, and (0) for the occupied lane of the reference vehicle.

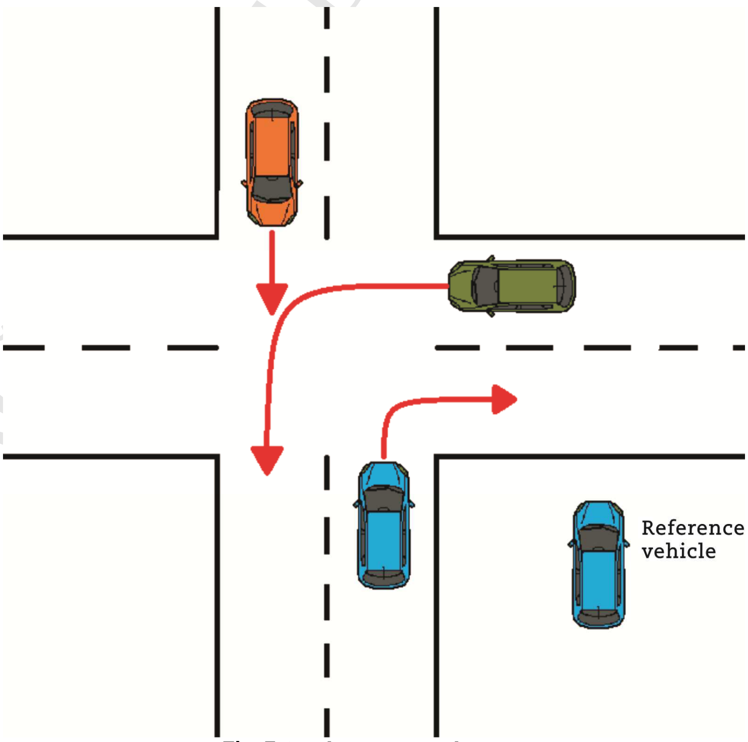

Fig. 7 Simple intersection diagram.

Alonso et al. (2011) theorized determining priority levels by twelve-point ordinal scale using, “very low, low, normal, high” with either a negative, no sign, or positive for priority placement. The ordinal scale references a look-up table for rules of setting priority, for example, a car turning right would be “high”. Vehicle communications between intersection participants would not be used to determine the CAVs priority level but to acquire the other vehicles’ intersection route plan. These two methods are a good base level control. Nonetheless, it did not discuss how intersections with multiple lanes in each direction would be handled.

Malikopoulos et al. (2018) offer a control solution that intends to optimize an intersection by minimizing the energy efficiency of CAVs and maximizing the throughput. They employ seven cases that rely on CAV’s controls, acceleration/deceleration capability, and the states relative to the intersection, position, speed, and time. Each case addresses how to handle passage through the intersection when neither controls or states are active, when all controls and states are active, or a varying combination of the two. It designates unique identities for every CAV that enters the control area of that intersection and registers the control and state cases of the vehicle. Simulations show an improvement of greater than 30% on reducing travel time and fuel consumption over standard traffic lights, but not all possible traffic routes were included in the simulation. There are opportunities for improvement with this solution: (i) vehicles, only of one size, were only permitted on straight through paths, omitting left and right turns in their proposal and (ii) a communication point monitors that intersection, but it does not calculate any crossing time or speed solutions.

Malikopoulos et al. (2018) expanded the work to consider left and right turns within the merge zone (Zhang et al., 2017). They also included a function of jerk in dealing with the comfort level of the passengers. The jerk function helps constrain the speed and acceleration while executing a turn in either direction. The research was built on three assumptions (Malikopoulos et al., 2018): 1) each vehicle i has proximity sensors and can observe and/or estimate local information that can be shared with other vehicles, 2) the decision of each vehicle i on whether a turn is to be made at the merge zone is known upon its entry in the control zone, and 3) for each vehicle i, the speed remains constant after the merge zone exit for at least a length δ (predetermined safety distance variable). As in study of Malikopoulos et al. (2018), the research also seeks to optimize the usage of fuel while traveling in the intersection regardless of path chosen. The trade-off between passenger comfort and fuel optimization is also considered. There are four general conditions, shown in Fig. 8, that occur at intersection: $\epsilon^z(t)$, $S^z(t)$, $L^z(t)$, and $O^z(t)$. $\epsilon^z(t)$ has CAVs from different lanes entering the merge zone converge on a single exit lane. $S^z(t)$ contains CAVs entering the merge zone on the same lane but exiting in various directions. $L^z(t)$ has vehicles entering the merge zone in multiple lanes and/or directions and exits the zone with paths that intersect one or multiple paths of other vehicles. $O^z(t)$ consists of CAVs that enter in various lanes and directions but do not have an intersection path with other vehicles in the same time period.

(a) (b) (c) (d)

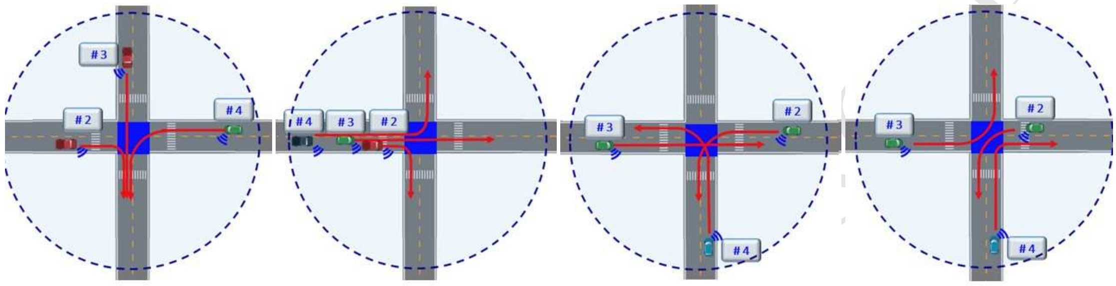

Fig. 8 Intersection conditions as used in (Zhang et al., 2017). (a) $\epsilon^z(t)$. (b) $S^z(t)$. (c) $L^z(t)$. (d) $O^z(t)$.

Using distributed conflict resolution, Liu et al. (2018) proposed a mechanism that increases throughput without the need of an intersection manager. The authors applied a cost function to give a weighted sum in setting an order of intersection occupancy while observing safety constraints. When a CAV calculates the intended time occupancy of the intersection, it broadcasts this data to affected participants. The other vehicles adjust their interception for the crossing using the same mechanism and broadcast updated data. The broadcasted information is used to determine trajectory calculation and passing order subject to temporal constraints. Trajectory calculations include spatial conflicts, tie breaking, and passing sequences. Temporal constraints are derived from an optimization problem formulated in study of Liu et al. (2017), where speed, acceleration, and jerk supply the limits to control the intersection time occupancy and passenger comfort.

Lots of traffic intersection control mechanisms have been developed for CAVs in recent years. We summarize the survey literature in Table 1. Centralized control often incurs high computational overhead, but it tends to be more reliable, especially when lots of CAVs are involved. Decentralized control, on the other hand, is more flexible and should be helpful when the infrastructure is unavailable. We think both types of control should be used in the field, and they may complement each other in big ways. In addition, it would be interesting to see the performance of a hybrid approach that combines both.

Table 1 Summary of results for intersection navigation.

Category	References
Real-world testing	(Wu et al., 2013), (Milanés et al., 2010)
Optimization and control	(Li and Wang, 2006), (Zohdy et al., 2012), (Lee and Park, 2012), (Lee et al., 2013), (Wu et al., 2013), (Kamal et al., 2015), (Levin et al., 2017), (Levin and Rey, 2017), (Malikopoulos et al., 2018), (Zhang et al., 2017), (Liu et al., 2018)
Heuristic	(Dresner and Stone, 2008), (Huang et al., 2012), (Zhang et al., 2015), (Wuthishuwong and Traechtler, 2017), (Lin et al., 2017), (Chai et al., 2017), (Colombo et al., 2017), (Wu et al., 2017), (Sarkar et al., 2017), (Milanés et al., 2010), (Alonso et al., 2011)
Isolated intersections	(Li and Wang, 2006), (Dresner and Stone, 2008), (Zohdy et al., 2012), (Lee and Park, 2012), (Wu et al., 2013), (Kamal et al., 2015), (Zhang et al., 2015), (Lin et al., 2017), (Chai et al., 2017), (Levin et al., 2017), (Levin and Rey, 2017), (Colombo et al., 2017), (Wu et al., 2017), (Sarkar et al., 2017), (Milanes et al., 2010), (Alonso et al., 2011), (Malikopoulos et al., 2018), (Zhang et al., 2017), (Liu et al., 2018)
Multiple intersections	(Lee et al., 2013), (Huang et al., 2012), (Wuthishuwong and Traechtler, 2017)
Environmental impacts	(Zohdy et al., 2012), (Lee and Park, 2012), (Lee et al., 2013), (Huang et al., 2012)
Human-driver consideration	(Dresner and Stone, 2008), (Lin et al., 2017), (Milanés et al., 2010)
Prioritized vehicles	(Zhang et al., 2015)
Pedestrian consideration	(Levin and Rey, 2017), (Sarkar et al., 2017)

Although these methods we review in this section show promising future of reducing traffic jam, fuel consumption, and emission, some gaps still exist, and more research efforts are needed. Most works consider CAVs only and do not consider manually driven cars. Given the fact that CAVs and human drivers will co-exist for some time, we need to consider how these two types of vehicles can share the traffic intersections efficiently. Most works assume that the information of the CAVs is readily available, but this assumption may not be always valid in real-world scenarios. Future research needs to derive the performance bounds when wireless packets are lost, interference is present, and transmission delay is non-trivial. This is especially important for decentralized control. Most works assume that cars are the only ones that pass-through intersections, but bikers and pedestrians also use intersections in reality. Future work should take bikers and pedestrians into consideration in the controller design. Most works assume regular intersections, and future work needs to be able to adapt to irregular ones easily. Most works do not consider traffic accidents, car breakdown events, power outages, inclement weather, and other factors that may suddenly happen. These incidents should be considered in future work to ensure safety and efficiency. Finally, most works are restricted to either isolated intersections or intersections in tandem We believe that it is important to jointly optimize a network of intersections in any topology to achieve maximum efficiency.

5. Collision avoidance

5.1. Maneuverability

One of the key aspects to the success of CAVs is the ability to avoid collisions with other vehicles. The primary option for avoiding most accidents is braking. However, in some scenarios, braking is not the best option or even a viable one due to high levels of traffic or the need to accelerate. A secondary method that needs to be considered is steering. Whether it is steering, braking, a combination of both, or some other means, a CAV needs to be able to respond to many circumstances where an accident may occur with the best countermeasure. This is where vehicle maneuverability can play a key role in the success of CAVs.

Current forms of autonomous collision avoidance have proven to increase the overall effectiveness of vehicle accident prevention. The Kusano et al. dive deep into several forms of autonomous collision avoidance methods to show the positive correlation between effectiveness and the increase of the number of collision avoidance systems used. In particular, using pre-collision system (PCS) algorithms for forward collision warning, pre-crash brake assist, and autonomous pre-crash braking, the authors simulate how crash speed is progressively reduced as each system is added on. Tests are run first with forward collision warning; compared with the systems without it, a 14% decrease in velocity change is observed (Kusano and Gabler, 2012). As pre-crash brake assist, and autonomous pre-crash braking are added on, respectively, the change in velocity continues to decrease by 10%, with each autonomous extension (Kusano and Gabler, 2012). The authors also note that in a real-world application, outside parameters (e.g., sensor limitation and weather) can affect the percentages.

Utilizing Sick LRS 1000, a laser scanner, Jimnez et al. (2014) designed a control system that consists of a single-layer infrastructure combined with time-to-collision (TTC) algorithms. The system determines when and how hard a vehicle needs to brake to avoid an accident. Additionally, if braking is not enough, the system takes control of (Kusano and Gabler, 2012) steering to maneuver the vehicle from crashing. The control system is set up into two levels: the first accounts for the vehicle's surroundings to determine viable open safe areas and the second computes the best way to evade an accident. The system functioned successfully in multiple controlled single-vehicle scenarios where the velocity is just above 30 km/h. Possible future research could be dedicated to the control system processing accident situations at much higher speeds and with multiple other vehicles involved in the surrounding area.

Predictive trajectory guidance (PTG) has the potential to be used as a means of transportation for semi and fully functional autonomous vehicles. By using MPC, an algorithm-based optimization method that adjusts outputs based on newly gathered input variables, PTG can be applied to ever-changing systems needed for successful autonomous driving (Weiskircher et al., 2017). Weiskircher et al. incorporate PTG into a close loop system that starts with navigation/GPS that transmits outputs to an assigner controller, which then sends a reference path as well as upper and lower control limits to the PTG. The PTG also receives information from the driver (primarily in the semi-autonomous case). Fig. 9 illustrates how the PTG outputs are transmitted to vehicle dynamics control and actuation level system at the end loop, which relays adjustment information back to the assigner through cameras, sensors, radar, etc. Weisskircher et al. are able to achieve success in simulations and hope that future work can be designated to real-world scenarios.

Braking and steering generally are the primary and secondary approaches utilized in autonomous accident avoidance research. However, a key tactic that commonly gets left out is acceleration. In some instances, the optimum method to avoid a collision is for the vehicle to accelerate (Brannstrom et al., 2010). Specifically, vehicles are treated as polygons whose outer perimeters are used to calculate the necessary procedures to avoid an accident. Although the research in this paper primarily applies to non-autonomous driving situations, the experimental prediction algorithms developed can be utilized to advance other autonomous algorithms, e.g., MPC. A few problems could arise with this work if applied to autonomous vehicles. First, the authors only apply steering, braking, and accelerating separately, but in some scenarios the best option is to apply two options. Secondly, because the research is not predominately focused on autonomous vehicles, it does not take in account the integration of sensors in autonomous control systems. Lastly, while the authors claim that the algorithms can be applied to real-time situations, no testing besides simulations has been run.

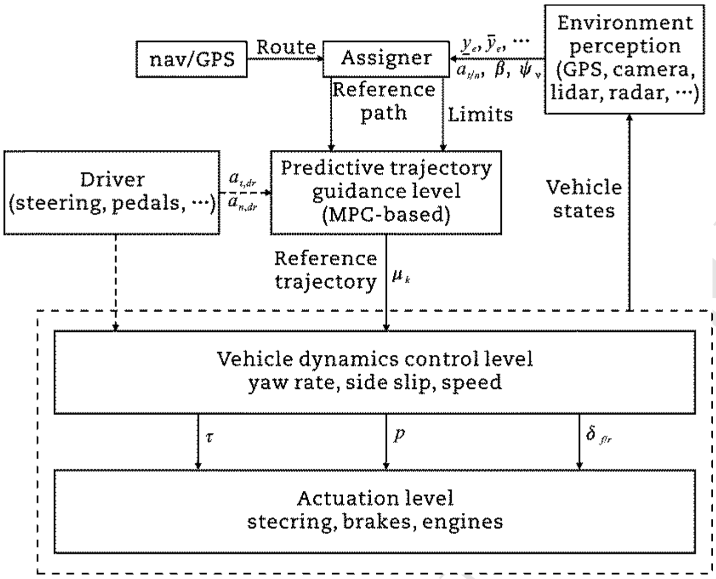

Fig. 9 MPC structural layout for semi-autonomous vehicles (Weiskircher et al., 2017).

The central focus of autonomous emergency braking generally deals with rear end collisions. However, the current design of many emergency braking systems does not take in account collisions involving intersections. Additionally, adaptive cruise control (ACC) capabilities are inadequate in extreme braking situation (Kaempchen et al., 2009). In this work, Kaempchen et al. concentrate on autonomous emergency braking (AEB) in both rear and intersection collisions with hopes of it potentially being implemented into ACC systems. The system uses a laser scanner to create two-layers collision prediction algorithms. The first layer is generated into simple preliminary algorithms. The system then quickly calculates a more complex second layer of algorithms. The two-layer algorithm setup allows the system to generate the algorithms with less complication. Fig. 10 is the setup of the ACC structure in study of Kaempchen et al. (2009). The system is designed to only activate when steering cannot avoid collision. Therefore, it eliminates false activation of the AEB, which could be instrumental in the success of autonomous vehicles. It also has the capacity to increase the overall smoothness of ride, reduce traffic build up, and provide safer driving for all other transportation around the vehicle.

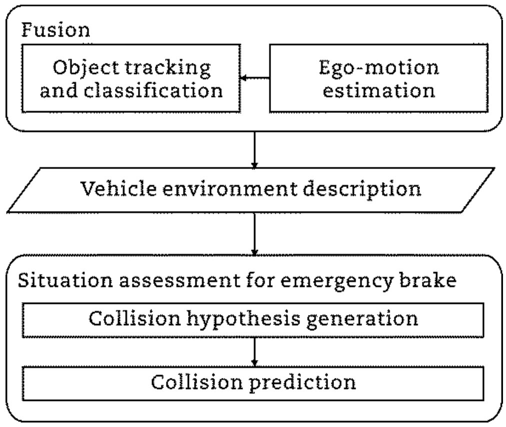

Fig. 10 System structure (Kaempchen et al. (2009)).

A critical hurdle for autonomous collision avoidance to overcome is finding a suitable balance between fluidity and overall safety. Sensor range and shortcomings tend to be the major contributing limitations to the overall effectiveness of collision avoidance systems (Nilsson et al., 2016), in which Nilsson et al. attempt to analyze these weaknesses in autonomous collision avoidance. They focus predominately on worst case scenarios to derive formulas from a brake threat number (BTN) and a steer threat number (STN). By doing so, they are able to analyze various collision avoidance scenarios: longitudinal, lateral, prediction error, and measurement. As verified by the simulation results, the derived theoretical formulas can be used to discover possible collision avoidance systems weaknesses and set sensor prerequisites.

5.2. Vehicle networking

Current forms of accident avoidance rely on the driver, sensors, and the vehicle's collision avoidance system. Nevertheless, the autonomous vehicles possess the potential to gain aid from other vehicles to avoid an accident. If a vehicle has no room to avoid a collision, the accident bound vehicle could potentially relay to another surrounding vehicle to adjust its speed and position. This will give more leeway towards the vehicle avoiding the accident. In general, through networking and estimation, autonomous vehicles can be systematized so that vehicles on the same network can assist one another in preventing collisions. Present studies have successfully run simulations, which test the effectiveness of networked autonomous systems with and without estimation (Moradi-Pari et al., 2016).

Networking can serve in more than one purpose for autonomous vehicle collision avoidance. If vehicles can communicate with one another, they can serve as a possible means to reduce single car accidents or multiple car collisions. Thus, for optimum results, collision avoidance should encompass individual autonomous systems and cooperative systems. Researchers have begun combining autonomous control system and machine learning algorithms to elevate the effectiveness of a potential dual system combining autonomous and cooperative systems (Desjardins and Chaib-Draa, 2011). Desjardins and Chaib-Draa direct their focus towards cooperative adaptive cruise control (CACC). They successfully engineered simulations, which were able to display the promise using autonomous and cooperative systems in CACC. Fig. 11 shows the structural breakdown of authors' concept of CACC. The information gathered from the environment is transmitted back to positioning system, which is passed back through so that the system may make the necessary adjustments. With some improvements in equipment and technology, there is a possibility that this can be applied to a physical application.

While vehicle networking can reinforce the safety measures with accident avoidance, vehicles' collision avoidance systems must still process the ability to perform without an accident avoidance network. Due to potential network failure and/or delays, collision avoidance systems should not be solely dependent on networking to function.

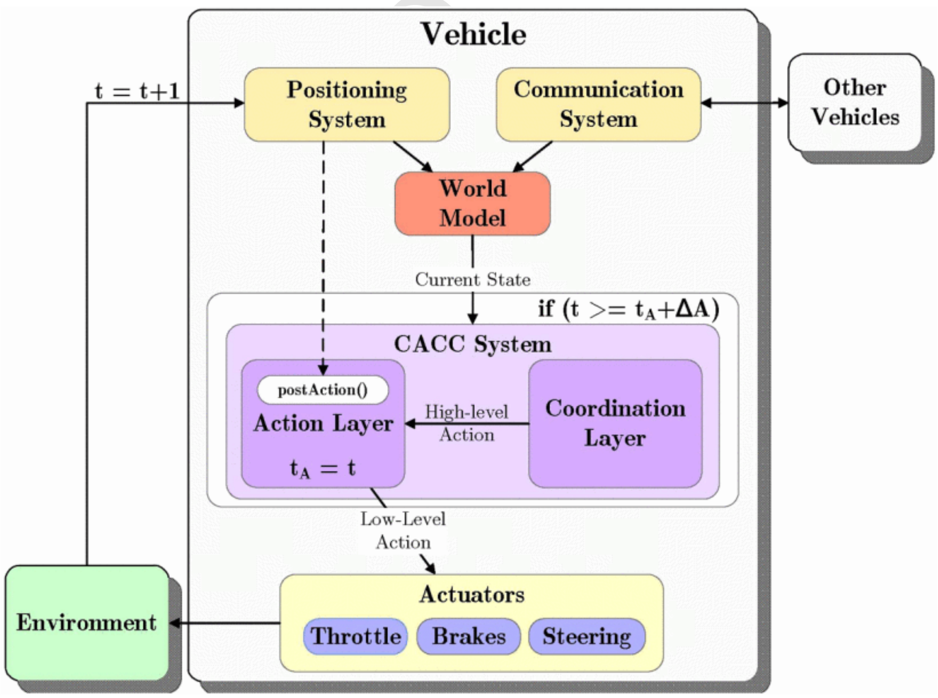

Fig. 11 CACC Layout (Desjardins and Chaib-Draa, 2011).

Nevertheless, it is beneficial that measures be put in place to booster the ability to find the root cause of a network failure or delay because networking enables collision avoidance to perform optimally. Elimelech et al. (2017) investigate ways to identify root cause failures in multi-agent systems, where resources are shared to coordinate positions for avoiding collision. They establish that methods can be incorporated into fail safe algorithms based on fault and conflict models generated from the analysis. In the future, work can focus on the generation of the algorithms that can locate network complications.

5.3. Control confliction

For collision avoidance to truly be successful in autonomous vehicles, it must be fully integrated with the vehicle's trajectory, counter-sway, and mapping systems; in addition, it should take precedence over them whenever needed. Countermeasures must be programmed in case of a potential conflict between another control system and the anti-collision system. For example, model predictive control (MPC) can be used to generate the essential control loop to allow the system to choose collision prevention over all other factors (Funke et al., 2017). Specifically, Funke et al. utilize Erlien et al.'s work (Erlien et al., 2013), involving vehicle path following and collision avoidance, to design their vehicle dynamics criteria and controller. They were able to create and run tests where an autonomous cart drove around a track with a designated trajectory. They then proceeded to add a pop-up obstacle in the path of the vehicle, resulting in a successful change in its direction to avoid it. The tests provide good scenarios to be applied to accident avoidance of autonomous vehicle and can provide the framework for more complex and dynamic collision scenarios, e.g. lane changing.

Lin et al. (2014) dive deeper and focus on collision avoidance in lane changing scenarios. The authors also add on DSRC and conflict probability to MPC, in order to further increase the vehicle's chances of success. DSRC adds another protective measure between multiple autonomous vehicles by allowing one vehicle to let another know that it is near before other sensors becoming active. Thus, the collision avoidance system (CAS) is designed to run checks through control loops to enable the vehicle to avoid a crash through lane-changing maneuvers only if there is not another vehicle in the adjacent lane. Lin et al. were able to run computer simulations of two-lane road scenarios successfully, but they have yet to apply it to a physical vehicle.

As technology improves with autonomous vehicles, the major obstacle of accident avoidance will inherently shift away from systematic shortcomings. Based off of data collected by Dixit et al. (2016), the second leading cause of autonomous disengagement was caused by driver initiation. Their research showed that the more miles traveled by a rider, the less likely they were to disengage the autonomous mode. However, there was a correlated increase in reaction time with the added miles. This brings up the dilemma of whether human or autonomous control should take precedence in accident scenarios. As reaction time continues to increase, the safer it becomes for the autonomous mode to remain in control. However, drivers might believe otherwise. The data also showed that the leading cause of deactivation of the autonomous mode was systematic failure. Thus, no matter how advance technology becomes, this will always pose a threat to accident avoidance, even if the possibility is extremely small. Therefore, sufficient safety measures must be in place to warn drivers in case of system failure.

Llorca et al. (2011) proposes that the CAS has complete superiority over the driver: in an emergency scenario, drivers have a high tendency not to perform the most efficient maneuver. According to study of Adarns and Place (1994), they are more inclined to brake over steer to avoid an accident or a pedestrian due to the fear of possibly causing an additional accident. Llorca et al. (2009) designed a system that would activate when a pedestrian was detected by an on-board stereo sensor. The vehicle was set up with a fuzzy control system that allowed pedestrian avoidance through steering maneuverability. Braking and outer car hood air bag were used as a last measure failsafe. The tests were run successfully in a traffic less scenario up to 30 mph. While more complex scenarios still need to be tested, Llorca et al. establish that one major advantage that CAS has over manual driving is the lack of distress, which can cause a driver to perform a non-ideal action.

5.4. Motorcycles

For autonomous vehicles to become mainstream, the accident avoidance rate must be far more efficient to that of manual driving. However, because the risk is far greater when motorcyclists are involved, the pressure for effective autonomous collision avoidance increases substantially. With the motorcycles' smaller frame and superior maneuverability, CAV's sensors need to have a wider range so that there are no blind spots where a motorcycle may enter.

Nevertheless, two-wheeled vehicles are making use of collision avoidance along with the four-wheeled counterparts. Two investigations took place in Europe called the powered two-wheeler integrated safety (PISa) and safety of powered two wheelers (CISAP). In the follow-up review (Savino et al., 2013), data were analyzed to see the possible outcome of fixing driver error and replacing inexperienced motorcycle riders with experienced ones in 37 past accident cases. Additionally, the effects of adding autonomous braking to the situations were examined. The two cross-examinations concluded that 67% of the cases would have benefited from the application of autonomous braking. In a similar study of the effects of two-wheeled autonomous braking, 100 accidents were simulated based off of real-world parameters (Savino et al., 2016). The results did not prove to be as confounding in showing the benefits of two wheeled autonomous braking as in study of Savino et al. (2013), but there still was a 14% improvement over manual braking.

Because of the ever-growing demand of innovative automotive collision avoidance, a large quantity of the research and development is being indirectly or directly dedicated to the field of CAV collision avoidance. Indirect measures that require various degrees of human control will have to be modified in some formats to avoid conflict between the driver and vehicle. However, with researchers developing their CAS to override human control in the time of emergency, this problem can easily be overcome. Even though currently developed measures of CAS prove to be more effective than non-autonomous methods, any errors in CAS prove to be more catastrophic due to the damage that occurs to CAV's appeal every time an incident takes place. Thus, CAS being superior to nonautonomous avoidance is not enough; in addition, CAS must be virtually perfect in order for it to catch on in today's automotive industry.

In addition, the social aspects need to be better understood and incorporated into CAS design. Suppose that a collision is unavoidable, what would the CAV do to minimize the damage? Questions like this need to be addressed in CAS design.

We also believe that CAS should be more cautious in situations such as inclement weather, stop-and-go traffic, and construction zone. Machine learning algorithms should be designed to provide warnings to CAS based on historical and current weather, location, and traffic data.

6. Pedestrian detection

The center for disease control and prevention discloses that over 180,000 non-fatal injuries were due to vehicles in 2015. Also, reports from the National Center for Statistics and Analysis reveal that there was a 9.5% increase in pedestrian-vehicle fatalities from 2014 to 2015 (NHTSA, 2017). Pedestrians pose the biggest hurdle to the success of CAS due to their high risk of injury and fatality when involved in vehicular accidents. CAS has been proven to aid in the performance of human drivers' ability to reduce pedestrian-vehicle collisions. With the continuous advances in autonomous collision avoidance and due to the limits in human capabilities, CAS can prove to be safer for pedestrians than human drivers altogether.

As can be seen in the collision avoidance section, braking is one of the principal autonomous accident prevention methods. Because AEB reacts 2 to 4 times faster than a human driver, the European New Car Assessment Program (NCAP) is implementing requirements that vehicles must have some form of AEB to receive high safety reviews. Currently, the primary AEB sensors being utilized to detect pedestrians are cameras, with radar sensors used as auxiliary aids. Cameras and short-range radars typically have the capabilities to identify pedestrians up to 60 m away, while long-range radars can go up to 180 m. Therefore, AEB reacts proportionally to a vehicle's speed, acceleration, and distance from a pedestrian (Park et al., 2017). Typically, AEB applies three warning zones that use an additional measure in each zone as the pedestrian gets closer, from a visible and audible warning to full brake activation. However, a key problem with current AEB systems is that they only take into account the longitudinal movement of pedestrians.

Park et al. (2017) set up a prototype AEB system that implements additional lateral movement aspect. Their system employs a funnel map system that takes into account of closest in-path pedestrian (CIPP) by creating two zones. The zones are broken down into a no-warning and a warning area: the former is 150% to 200% of the vehicle's total width, and the latter is the width of the vehicle. Fig. 12 shows funnel mapping pedestrian selection (Park et al., 2017). This allows for the system to recognize which pedestrian is the CIPP and react to a potential change in the CIPP if a pedestrian leaves or enters the warning zone. Additionally, it provides the AEB with a prediction (no-warning) zone, which gives the CAV a faster reaction time to pedestrians. Park et al. also developed distance formulas that correlate to speed, time, and acceleration that correspond to different zones. It was determined that for speeds above 50 km/h, steering proves to be a better means of avoidance for pedestrians than braking, while braking is more efficient for speeds below 50 km/h (Park et al., 2017). When tests were run for their AEB system against the standard, it reduced braking speeds by somewhere between 7.4 and 11 km/h and increased braking distance by somewhere between 1.1 and 1.5 m. Funnel mapping's warning zones can prove to be highly valuable for pedestrian detection, especially because studies have shown that pedestrians are more inclined to distractions than drivers (Rangesh and Trivedi, 2018), nearly 30% of crossing pedestrians undertake in distracting actions. Thus, Park's setup of funnel mapping accommodates for distracted pedestrians without creating unnecessary braking for pedestrians at risk. However, even with the advantages of the funnel mapping systems, many problems remain with camera sensors' visual detection capabilities, which decrease in inclement weather.

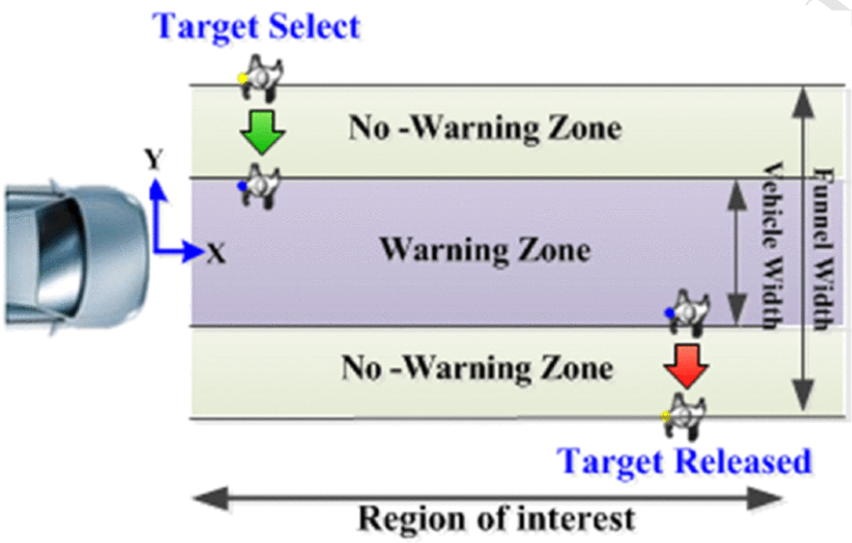

Fig. 12 Funnel mapping diagram (Park et al., 2017).

With the recent influx of technological advances in machine learning and computer vision, a crossover with some of these techniques into CAVs can prove to be beneficial. Dominguez-Sanchez et al. demonstrate how convolutional neural networks (CNNs), a form of machine learning, can be used to decide the necessary vehicular action based on pedestrian movement predictions (Dominguez-Sanchez et al., 2017). CNNs initiate with an input layer based on some form of imaging, and they use intertwined mathematical algorithms to generate a predictive output layer. Dominguez-Sanchez et al. (2017) use a high capacity to test three different CNN architectures (AlexNet, GoogleNet, and ResNet) and their motion prediction capabilities of sequencing frames. Overall, the results confirmed that the ResNet has the most superior recognition performance with an accuracy in the validation of 94%. Further testing will be necessary to show the effectiveness of CNNs in a moving vehicle, but Dominguez-Sanchez et al. have proven that CNNs have the capacity to be incorporated into CAS.

The further ranges of obstacles or pedestrians are from vehicles, the less effectiveness of sensors, radar, and detection capabilities becomes. Inclement weather and low light situations amplify the deterioration of their performance. Fig. 13 shows the decline of the LIDAR cloud density as the range of a pedestrian increases. Increasing the range and power of these variables is a requirement for pedestrian detection in CAV applications. Li et al. (2016) design a density enhancement method, which can be used to bolster pedestrian detection at further ranges. In this work, a three-step density enhancement process was developed to be incorporated in between the segmentation and feature extraction steps of 3D LIDAR pedestrian detection. The enhancement starts with devising a local coordinate system, followed by performing RBF-based interpolation and ending with resampling. The RBF-based interpolation integrates cross-validation with an error metric, which allows the density enhancement to form parameters and a kernel. The resampling allows for the method to take a few captured points and transfer them into a binary pedestrian format. Note that the density enhancement focuses on stationary pedestrians only. With future work, the method could be combined with funnel mapping to further aid in the advancement of CAV pedestrian detection.

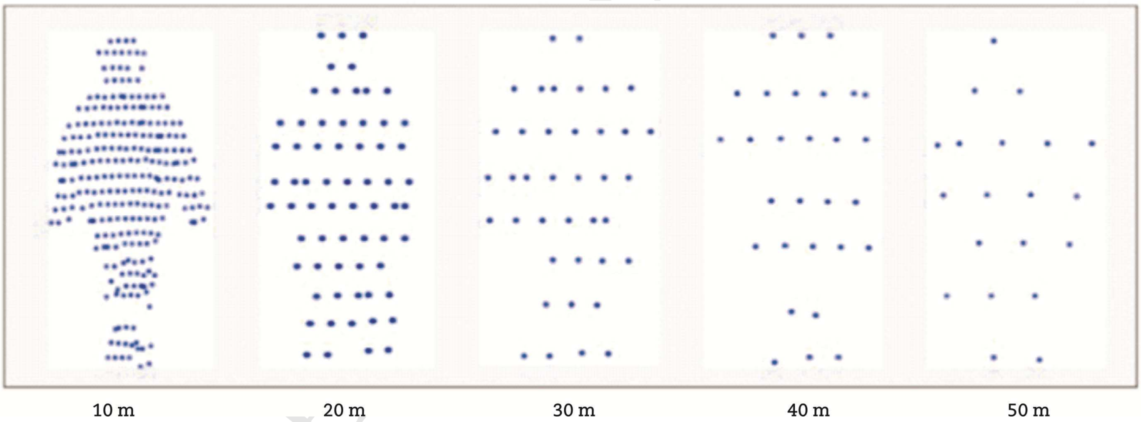

Fig. 13 Cloud density decline (Li et al., 2016).

Pedestrian detection is one of the most intricate tasks for CAV technology to overcome. While basic pedestrian motion can be depicted into simplified rudimentary dynamics, pedestrians' decisions and actions can be quite erratic and unpredictable, and it poses an obstacle. Thus, CAV pedestrian detection must accommodate the many possibilities of pedestrian motion. Xiao et al. (2017) refer to the importance of simultaneous localization and mapping (SLAM), which customarily deals with only static scenarios (Bresson et al., 2017; Engel et al., 2013; Klein and Murray, 2007; Mur-Artal et al., 2015; Newcombe et al., 2011). As the name implies, SLAM is a trajectory optimizing navigation network, which concurrently adapts the vehicle's path based on obstacles that may appear in study of Xiao et al. (2017). Xiao et al. attempt to topple the task of combining static and dynamic object detection with CAVs, and an algorithm was developed to hone rough object detection into a smooth output. Specifically, the system generates super-pixels from a disparity map and optical flow field. From there, a two-stage conditional random field (CRF) model first establishes the static objects, followed by indicating the dynamic targets; the final stage is the fine tuning of the dynamic segmentation and potential motion estimation. To test the performance of the algorithms, MATLAB simulation in junction with the KITTI dataset (Geiger et al., 2012, 2013) are used.

While inclement weather poses a hurdle for pedestrian detection to overcome, a more common obstacle for pedestrian detection is nighttime identification. Choi et al. (2018) note that most datasets, including KITTI (Geiger et al., 2012, 2013) and Cityscapes (Cordts et al., 2016), tend to only perform efficiently in bright environments; thus, it is priority for autonomous pedestrian detection that improvements need to be made to the capacity of datasets if they are going to be a primary form of recognition. The KAIST multi-spectral dataset (Choi et al., 2018) has the potential to provide the framework to solve the shortcomings of other datasets. KAIST is a broader

scale dataset designed to function during the day as well as nighttime and dim light scenarios. To achieve the goals, Choi et al. use a beam splitter, an optical device that operates within the visible and thermal spectrum and fuses the results of RGB and thermal imaging. With the addition of LIDAR, their system eliminates various lighting complications. Sensor and camera calibration play a much greater role in the work of Choi et al., due to the multi-function capabilities of KAIST and the usage of beam splitter. Therefore, the system requires more frequent calibrations than other dataset systems.

From the statistical data, pedestrian detection poses one of the biggest challenges for both non-autonomous and autonomous vehicles. Nonetheless, a wide range of technological methods can be applied to CAV pedestrian detection. Thus, it provides an inlet for researchers from different disciplines to work together towards the success of CAVs. The large-scale success of this CAV feature will require more than one discipline to thrive in employing its developments. Researchers must continue to improve on ways to connect their work through CAV pedestrian detection so that it may be implemented further than just theory or prototypes. There must first be more supplemental work dedicated to real-world emulation, followed by physical implementation.

Simply letting the CAVs detect the pedestrians may not be enough. We believe that the pedestrian detection systems equipped on CAVs should be smart enough to understand body languages and hand gestures, which have been the primary way of communication between human drivers and pedestrians at crosswalks. Artificial Intelligence needs to be developed in order to understand the intention of pedestrians so as not to create unnecessary slow down and accidents.

In addition to identifying pedestrians, CAVs also need to be able to recognize animals such as deer, dogs, and kangaroos. Animals may not create problems in cities, but they cause lots of traffic incidents in rural areas. The ability of animal detection will greatly ensure the safety of CAVs and their passengers.

7. Conclusions

Within the past decade, the advancement of technology that is built into or applies to CAVs is improving at a significant pace. While recent studies and experiments that we have discussed show success, the next step will be to integrate the components into a single platform without security or operating conflicts. Combining collision and pedestrian avoidance into the intersection navigation control without elevated risk will be a major hurdle for the general population's safety. The opportunities for incidents reside in the urban intersection with large amounts of pedestrian and cycling traffic along with human controlled vehicles and CAVs. It is imperative that each component needs to work flawlessly without slowing down or restricting the overall throughput of intersections and general roadways. Inter-CAV communication is the gateway that RSUs, intersection managers, CAVs, and other safety devices use to alert each other and pedestrians. Of the topics that have been discussed, collision and pedestrian avoidance is near the apex of importance in the need for preservation of human life.

The future opportunities for technologies discussed in this paper include increasing the overall effective range of V2X communications from the approximate 500 feet radius and further congestion control in message rates to keep the delay of sent messages to a minimum. In addition, the sensor resolution used in LIDAR, RADAR, and others is affected by inclement weather. Reducing the impact of the weather to signal to noise ratio found in these sensors, for better distinction of vehicles or pedestrians, is an important prospect; the need for higher resolution permits decreased reaction time and increased overall performance for the CAV, especially in inclement weather where every millisecond counts. Furthermore, the work in centralized intersection control allows for safe hand-offs of route parameters, but decentralized intersection control bypasses the need for exterior path calculations. We believe that continuing optimization of decentralized intersection control mechanisms will keep the total investment for the general population lower than compared to a centralized option.

Conflict of interest

The authors do not have any conflict of interest with other entities or researchers.

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